

ing in reduced electricity consumption and a more sustainable workspace.

Consumer electronics. Indoor solar cells offer a sustainable alternative to disposable batteries for consumer electronics, including remote controls and wireless keyboards.

Navigating challenges to unlock full potential

Due to reduced light intensity and different spectral compositions, indoor solar cells generally have lower efficiency than outdoor cells. However, advances in material science and cell design have improved their performance. Outdoor solar cells, such as crystalline silicon cells, typically achieve 15% to 22% efficiency under optimal sunlight. In contrast, indoor solar cells like amorphous silicon and organic photovoltaic cells exhibit 10% to 15% efficiency under indoor lighting, with dye-sensitised and perovskite-based cells showing potential for even higher efficiencies in controlled environments.

Efficiency challenges. Indoor solar cells face significant efficiency challenges compared to outdoor cells in low-light conditions. Traditional solar cells, optimised for sunlight, perform poorly indoors. Researchers are improving efficiency by developing materials like organic photovoltaic and perovskite solar cells and advancing light-harvesting techniques and cell designs. However, achieving high efficiency remains a significant challenge for widespread adoption.

Cost. Compared to traditional energy sources, indoor solar cells can be costly due to expensive materials and manufacturing processes for advanced technologies like organic photovoltaic and perovskite cells. Integrating these cells into products adds to the cost. Although prices are decreasing due to economies of scale and technological advancements, they remain more expensive upfront than grid electricity or batteries. Long-term energy savings and environmental benefits may offset

initial costs. Incentives, subsidies, and manufacturing advances are crucial for making indoor solar cells more economically viable.

Adoption barriers. Several barriers hinder the adoption of indoor solar cells. Consumer awareness and acceptance could be higher, and integrating solar cells into existing products requires significant design and manufacturing changes. Reliability and performance concerns also deter potential users. Solutions include increasing public awareness through education and marketing, offering financial incentives and subsidies, demonstrating reliability through pilot projects and case studies, and developing standardisation and certification processes to build trust and facilitate integration.

Design and aesthetic considerations

Indoor solar cells can seamlessly blend into household items like window blinds, light fixtures, furniture, and decorative panels, collecting energy without compromising the aesthetic of living spaces. They can be embedded in translucent surfaces, remaining virtually invisible.

These solar cells come in various sizes, shapes, and colours to match different interior decor styles. Their versatility allows integration into diverse items, from small electronics to large architectural features, accommodating specific design needs such as curved surfaces or patterned finishes. This ensures they meet both functional and aesthetic demands for energy-efficient interiors.

Pioneering the future

Ongoing research and development in indoor solar cells focus on overcoming limitations and unlocking new possibilities through advanced materials like organic photovoltaics and perovskites. Research institutions and companies are dedicated to enhancing performance, reducing costs,

and improving durability and lifespan. Collaboration between academia, industry, and government accelerates innovation and commercialisation, with pilot projects and real-world demonstrations providing crucial data for further advancements.

Leading laboratories worldwide are exploring novel materials to increase the efficiency and durability of indoor solar cells, solidifying their position as a viable alternative for sustainable energy solutions. Researchers are developing seamless integration techniques that allow solar cells to be embedded into various surfaces and objects, enhancing both practicality and aesthetic appeal. A key area of focus is on advanced energy harvesting systems, which are designed to store and manage power more effectively, ensuring a consistent energy supply. Improving the durability and longevity of indoor solar cells is also crucial, making them more cost-effective over time.

Innovations in material optimisation, integration techniques, energy harvesting systems, and durability are essential to establishing indoor solar cells as a mainstream energy solution. These advancements hold the potential to revolutionise how energy is harnessed in various settings, paving the way for a more sustainable future.

The rise of indoor solar cells is poised to power the future, boosting energy efficiency, reducing reliance on traditional power sources, and supporting sustainable development. They offer a versatile and eco-friendly energy solution for smart homes, offices, industrial applications, and consumer electronics. With continued innovation and adoption, indoor solar cells are set to play a crucial role in achieving a sustainable future. **EFY**

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